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Cyber threat detection: Unsupervised hunting of anomalous commands (UHAC)

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Highlights

- Threat hunting is searching through logs to locate malicious activities.
- We propose a novel method, UHAC, that can perform anomaly-based threat hunting in SIEM logs for process auditing.
- The UHAC detector identifies 84–89% of anomalies when the top 10% of the data is targeted.
- The detector consistently outperforms other algorithms in detecting anomalies.

Abstract

The cyber security industry is rapidly adopting threat hunting as a proactive tool for early and faster detection of suspected malicious actors. In this paper, we propose a machine learning-based method, Unsupervised Hunting of Anomalous Commands (UHAC), to detect text-based anomalous commands in security information and event management (SIEM) logs that are good candidates for

threat hunting. A unique feature of the proposed method is that it first creates a feature set based on the augmentation of document-term and document-character matrices. Then, an autoencoder-based detector is trained on this feature set using a custom loss function. UHAC consistently outperforms other feature sets and algorithms such as one-class support vector machine, density-based spatial clustering of applications with noise, and word-embedding based models such as *word2vec*. The UHAC detector identifies 84–89% of anomalies in the top 10% of the data. Findings have implications for cybersecurity analysts who perform threat hunting in SIEM logs for process auditing on endpoint devices.

Introduction

The global market for cyber security solutions and services was valued at approximately \$217 billion in 2021 and is expected to grow at about 10% annually to reach \$345 billion by 2026 [1]. This rapid growth is in response to the increasing sophistication of threats capable of evading first lines of defense such as endpoint protection or firewalls [2].

An emerging approach that helps keep malicious actors at bay is *threat hunting*, which involves *searching through system logs to locate malicious activities that evade existing security solutions* [3,4]. As a proactive approach, threat hunting reduces the time it takes to detect threats that penetrate organizations [5]. In fact, 91% of organizations that employ threat hunting report benefits that range from faster detection of threats to preventing damage [6].

There are different types of threat hunting. Anomaly-based threat hunting involves identifying anomalies in logs generated by security information and event management (SIEM) tools [5]. Note that SIEM tools not only aggregate alerts, reports, and logs collected from organizational devices, but also identify suspicious activities by applying pre-defined rules [7]. Therefore, logs captured by SIEM tools are already labeled as suspicious. However, identifying anomalies in these logs is not easy because they contain millions of records, 50% of which are false positives [8]. Therefore, more than 50% of legitimate threats usually are undetected [9]. The ability of SIEM tools to conduct *process audits*, which involve capturing every command executed on each endpoint device, exacerbates the situation. Accordingly, SIEM tools capture commands executed by both users and applications [10,11]. Because they are text based and unstructured, cybersecurity analysts might not be able to systematically determine if these commands pose any threats. If there are too many false positives, threat hunting becomes economically unviable because of its challenging and time-consuming nature [12,13]. Therefore, analysts need a decision support system that can help them identify anomalous commands in SIEM logs so that they can focus on them to determine their level of threat. Note that an anomaly might not pose a threat or be malicious. However, an anomalous command indicates that a process that is different than others is executed on a device. Therefore, this command should be identified and presented to an analyst to determine whether it poses a threat.

Prior work has used supervised models in the cybersecurity domain. These models require labeled data, which is very difficult, if not impossible, to generate given that the emergent cybersecurity threats are often unknown and data sets are very large with millions of records that vary from one context to another. Therefore, in this paper, we propose an unsupervised method to address these challenges. Specifically, we propose a method, hereafter referred to as “Unsupervised Hunting of Anomalous Commands” (UHAC), that can detect anomalies in a large corpus of text-based commands to help analysts automate threat hunting in SIEM logs for process auditing. Using this method, we propose a UHAC detector that identifies anomalies without the need to label data or create a training data set for only normal commands. Note that in the context of this study an anomaly refers to a unique (or infrequently occurring) command compared to all other commands.

A unique feature of the UHAC detector is that it parses text-based commands at both term and character levels. Term-level parsing generates a document-term matrix with binary weights. Note that terms are identified using a custom parser built from a natural language processing library. Character-level parsing generates a document-character matrix with custom weights based on character frequency-inverse document frequency. Characters are identified using the built-in parser of the Scikit Learn library. Note that terms and characters are identified organically from the data set without the use of a predefined vocabulary or alphabet. We create a feature set by combining these document-term and document-character matrices using matrix augmentation. As we demonstrate, this feature set significantly improves anomaly detection compared to term-level parsing, which is one of the most common feature sets in text mining. Further, we create this feature set without relying on complex or rule-based templates employed in earlier work e.g., [14,15,16].

Our work offers several advantages over existing approaches. First, we do not require a labeled data set or a dataset consisting of only normal events. Instead, we analyze the entire dataset and identify anomalies from within the same dataset. This makes our work truly unsupervised. Also, we address a critical limitation because existing approaches assume normal behavior is known upfront e.g., [14,16,17,18,19]. Second, whereas most of the existing approaches perform anomaly detection in the logs of high-performance computing (HPC) technologies e.g., [[14], [15], [16],18,19,20], our approach can be used to detect anomalies in any type of text-based data set. This is because we do not rely on predefined and handcrafted templates that are used to find patterns in HPC logs. Third, we examine individual commands that make up the processes executed on a device. Prior work uses either the metadata of processes e.g., [17,20] or their outputs e.g., [[14], [15], [16],19,21] to identify threats. Therefore, rather than relying on the byproducts of processes, our work focuses on actual processes.

To determine the effectiveness of UHAC, we conduct a series of tests. First, we show that the proposed feature set performs better than alternative feature sets commonly used in extant literature. Second, we show that the proposed UHAC detector outperforms some of the state-of-the-art techniques such as one-class support vector machine (OC-SVM), *word2vec*, and density-based

spatial clustering of applications with noise (DBSCAN) even when they use the proposed feature set. Third, we demonstrate that the results of our work are repeatable on a new data set.

Our work makes important contributions to the literature. First, following the footsteps of earlier work see [22,23,24], we leverage ordinary feature sets to propose a novel feature set. By augmenting the bag-of-words-based feature set with a custom bag-of-characters-based feature set, we show that we can perform better anomaly detection than using either of the two approaches individually. Therefore, we contribute to extant work by extracting more features from unstructured data sets and capturing better representations of text-based data. Second, we propose a new loss function for training autoencoders. The need for this loss function arises from the feature set's hybrid data type. We conduct robustness tests and demonstrate that our proposed loss function outperforms traditional loss functions such as mean squared error and binary cross-entropy. Third, we propose a novel method for hunting threats in SIEM logs for process auditing. This addresses a critical need in the field of cybersecurity where there is a dearth of decision support systems that leverage analytics and perform risk assessment see [25,26]. We believe our method can spur research in this area so that cybersecurity analysts can use better tools to find threats, and thus, prevent malicious acts.

Our work has important implications for practice as well. First, we show that our method can identify anomalous, or unique, commands captured in SIEM logs for process auditing. This can reduce analysts' time significantly in threat hunting and help them conduct investigations. Second, we show that threat hunting can be performed using unsupervised learning without the need to create labeled datasets. This is important because the definition of normal might change over time. Approaches that do not account for changes are likely to generate more false positives, reducing their viability in hunting for threats. Third, our method provides analysts with each observation's probability of being an anomaly. This is a departure from existing tools in the literature that only determine whether an observation is an anomaly or not using a binary classification. Using this probability, our detector sorts observations by their likelihood of being an anomaly, thereby enabling analysts to determine how deep of an investigation they want to conduct. As a result, analysts can identify hard-to-detect false negatives as they delve deeper into the results of the UHAC detector.

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Section snippets

Background and prior work

Threat hunting can be defined as *searching through networks and system logs to locate malicious activities that evade existing security solutions* [3,4]. There are at least three types of threat hunting [5]. *Intel-based* threat hunting involves obtaining indicators of compromise, such as hash values, IP addresses, and domain names, from intelligence sources and searching for them in SIEM logs. *Hypothesis-based* threat hunting concerns determining whether the tactics, techniques, and procedures ...

Unsupervised hunting of anomalous commands (UHAC)

To address the gaps and limitations described in the previous section, we propose a new method, referred to as UHAC hereafter. Details of this method is presented in Fig. 1. As shown in the figure, a SIEM tool aggregates the process auditing logs of each endpoint protection technology into a repository. We use this repository and extract the command associated with each process. First, we run these commands through a preliminary filter to identify all single-term commands. We do this because, ...

Data set

We obtained a data set exclusively from a cybersecurity firm that operates a security operations center (SOC). The firm has global presence in three continents and serves clients across the world. The data set includes the process audits (i.e., SIEM logs) of an organization captured for three weeks by a popular Endpoint Threat Detection and Response (ETDR) tool. Due to privacy concerns, the firm removed all personally identifiable and confidential information from the data. Even though data was ...

Evaluation

We test the goodness of UHAC using the *cumulative gain chart*, which determines the percentage of anomalies that are identified when a specific percentage of the data is targeted. To do this, we reconstruct each command using our UHAC detector and sort all commands by the reconstruction error from highest to lowest. Because anomalies are likely to have high reconstruction errors, the cumulative gain chart helps us determine how many anomalies are identified in each percentile of the data set. ...

Experiments

The analysis data set had a total of 8708 terms (i.e., vocabulary). We used only 1 % (or 87) of these to create the truncated document-term matrix. Also, there was a total of 47 characters to create the

document-character matrix. This led to a combined feature set with 134 features. We trained the UHAC detector to reconstruct these features. To ensure the reliability of results, we built five detectors simultaneously and calculated the average reconstruction error for each observation. Then, we ...

Preliminary filter

In the final version of the analysis data set (consisting of 10,485 events), the preliminary filter identified a total of 141 observations that consisted of a single-term command. Four of these were injected anomalies while the remaining 137 were normal observations. All 141 observations were removed from the analysis data set because, as discussed earlier, a frequency table can identify the number of times these observations occur in the data set. Thus, least frequently occurring commands can ...

Discussion

In this paper, we propose a novel decision support system to automate threat hunting among SIEM logs geared toward process auditing. Because process auditing involves logging the commands of all processes spawned on endpoint devices, the method allows cybersecurity analysts to identify unique, or anomalous, commands so they can conduct further investigation.

Our method has several steps. First, we perform data cleaning by removing all punctuation characters from commands. Then, we parse the ...

Conclusion

In this paper, we propose the UHAC method to perform anomaly-based unsupervised threat hunting in SIEM logs collected for process auditing. The method relies on a novel feature set based on a truncated document-term matrix with binary weights augmented with a document-character matrix with CF-IDF weights. The algorithm used for the method is an autoencoder that reconstructs this feature set using a custom loss function. The evaluation of this method is performed by cumulative gain chart, which ...

CRedit authorship contribution statement

Varol O. Kayhan: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing. **Manish Agrawal:** Conceptualization, Writing – review & editing. **Shivendu Shivendu:** Conceptualization, Writing – review & editing. ...

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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...This is demonstrated by achieving over an 80% true positive rate and true negative rate, alongside an F3-score exceeding 60%. UHAC (Kayhan et al., 2023): This study introduces the unsupervised hunting of anomalous commands (UHAC) system, which uses an unsupervised ML method designed to detect anomalous commands within SIEM logs. UHAC leverages the power of autoencoders trained on a unique combined feature set derived from both document-term and document-character matrices....

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Manish Agrawal teaches courses in information assurance, business data communications and applications development. A professor in and former chair of the School of Information Systems and Management, Agrawal has been the recipient of USF's university-wide award recognizing teaching excellence. An expert in the areas of cyber security, social media analytics, software quality, offshoring and outsourcing, his research interests include extreme event response, social media analytics, decision fusion and software quality. He is an avid researcher and his work has been published in numerous academic journals, including Management Science, MIS Quarterly, the INFORMS Journal on Computing, the Journal of Management Information Systems, IEEE Transactions on Software Engineering, Decision Support Systems and the Journal of Organizational Computing and Electronic Commerce. His research and teaching have been funded by the U.S. National Science Foundation, the U.S. Department of Justice, the Indo-U.S. Science and Technology Forum and Sun Microsystems. Prior to joining academia, Agrawal was a member of the Indian Police Service. Agrawal earned a PhD in information systems at SUNY Buffalo and studied at the Indian Institute of Technology in Kanpur, India.

Shivendu Shivendu is an associate professor of information systems. He has created and taught undergraduate, MBA, master's and doctoral courses in areas related to economics of information systems, blockchain technology, FinTech, IT strategy and the design of information systems. He is the principal investigator on the prestigious White House initiated Federal TechHire grant of \$600,000. He is the co-organizer of the annual "State of the Region" conference held Tampa, an event that draws hundreds to learn more about how the region compares to other metropolitan areas in key areas such as unemployment, income inequality, economic growth and educational attainment. Shivendu's research focuses on the emerging business and policy issues at the intersection of technology, economics and public policy. He employs analytical, empirical and experimental methods to study technology mediated business models, economics of digitization of information, pricing of digital goods including content pricing in dual mediums, sourcing of IT services, security and privacy in big data, business challenges of cloud adoption, health IT and technology policy. His work has been published in information systems journals including Management Science, Information Systems Research, the Journal of Management Information Systems, and Information Technology Management. His research papers have won many best-paper awards in conferences. He earned a PhD and a master's degree in economics from University of Southern California, Los Angeles, an MBA from the Indian Institute of Management, Ahmedabad; and a bachelor's degree in electrical engineering from the Indian Institute of Technology, Kanpur.

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